

# ABRA — the plain-English deck

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A slide-by-slide, jargon-light tour. The white paper (linked on the last slide) has the math and sources.

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## Slide 1 — ABRA

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A family of small AI models for competitive Pokémon (Champions VGC). Built from thousands of real ladder games. Honest about what it can and can't know.

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## Slide 2 — The big idea

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You **can't reliably predict who wins** a match from the two team sheets — in this format it's close to a coin flip, and even a strong player-rating model barely beats a coin. So ABRA stops trying to call the winner and instead **helps you make better decisions**: what to bring, and what to do turn by turn.

This is the same move sports analytics made with "expected goals" — you can't predict the final score, but you can measure the value of each shot. ABRA is that idea, pointed at Pokémon.

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## Slide 3 — How it's built

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One rule: **keep every real game, and analyse on top of it**. ABRA saves every public ladder replay into a growing library and builds its models on that library — so it gets smarter as more games come in, and never has to re-download anything. It runs on a normal laptop; no special hardware.

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## Slide 4 — The town of models

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Each model is a "house" you can visit on the site:

- **MEDICHAM** — the damage engine. Matches the community standard, but disagrees with the game's OFFICIAL engine by a wide margin, so it is being replaced by that engine and kept only as a lookup.
  - **MEW** — plays the official engine against itself, so we are not limited to the games people happen to upload.
  - **GURU** — reads the metagame: which team styles beat which, from real results, with error bars.
  - **XATU** — reads the opponent: the likely item, ability, and moves behind each Pokémon.
  - **PORY** — mid-battle win chance. **Retracted as a finding**: it works out to counting how many Pokémon each side has left and how healthy they are, and it does not beat exactly that baseline.
  - **The scoring bot** — the part that finally *looks at the board*.
  - **SLOWKING** — the strategist: there's no single best team, so it plays a smart mix.
  - **CHOMP** — the team picker: which four to bring, which two to lead.
  - **ALAKAZAM** — the coach (in progress): the best move to make, right now.
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## Slide 5 — The win: the practice opponent can finally see

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Everything ABRA claims about "this build beats that build" comes from playing the game out against a practice opponent. That opponent used to pick its move purely by **how popular the move is** — it had no idea what was standing in front of it. So it fired attacks at Pokémon that were flat immune, it used moves that couldn't possibly work, and when there were two enemies to aim at, it picked one by flipping a coin.

We fixed it by **watching what people actually do**. We took about 48,000 real decisions from games where both players' full teams were public — so we could see not just what someone clicked, but everything they *could* have clicked — and learned how much a real player's choice depends on what's in front of them. Nobody wrote the rules; they were measured.

The result, checked on games the learning never saw:

- it now hits the enemy's weakness **half again as often** as before,
- it wastes **a third fewer** moves on things that simply cannot work,
- and it almost never fires at something immune any more.

It is still short of a human on all three, and one thing it still does badly is decide **when to switch** — that's the next job. But the practice opponent is no longer playing blind, which means the numbers it produces are worth more than they were.

**PORY** also beats a coin flip mid-game, using how many Pokémon each side has left and their health, and it's *calibrated* — when it says 70% it's really about 70%. It's live on the site as a per-turn "you're at X%".

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## Slide 6 — The honest parts (this is a feature)

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Two things we tested and reported straight, even though they're negatives:

- **Picking the team doesn't beat a coin.** We tested whether the team-picker's choices track who wins — they don't, more than chance. So we don't oversell it. The damage math is still exact and useful.
- **The "rock-paper-scissors" metagame is a hint, not a fact.** The data suggests Trick Room beats Hyper Offense beats Sand beats Trick Room — but each of those rests on only ~13–18 games, so the error bars are wide. We label it suggestive, and it'll firm up as more games arrive.

Saying what we *can't* prove, as plainly as what we can, is the whole point.

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## Slide 7 — What's next

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**The one lesson that changed the plan.** We spent a day teaching the bot more about Pokémon — four separate things it didn't know before. All four made no difference at all. Then we changed *what it was trying to do*, twice, and both times it got a lot better.

Teaching it more facts had stopped helping. Changing its goal was what worked.

The reason is simple. The bot was trained to **copy what humans click**. That is a different goal from **winning**. It even shows up in the numbers: the model rates "use a spread move that hits both of them and doesn't hurt my own partner" as a *bad* idea — not because it is bad, but because humans rarely click it. A bot told to avoid its own best plays will avoid them.

- **Teach the two Pokémon to act as a team.** We already built this once. It plays worse, and now we know why: it was taught to *resemble* people rather than to *win*. Retraining it on the right goal is the single most promising thing on the list.

- **Fix a small unfairness in the training data.** When a Pokémon is locked into one move, we were still showing the bot all four as if it had a choice. It affects about 1 in 15 items.
- **Find out how readable it is.** A rival bot built only to beat ours won 63% of the time. That was measured a long time ago and never repeated. A bot can get better on average while staying just as easy to read — those are different questions, and only one of them has been checked lately.
- **Then, looking ahead.** Right now the bot only thinks about *this* turn. Looking one turn further is a real project, but it's now measured rather than guessed: there are only **72** sensible combinations to consider each turn, not the trillions people assume.
- **ALAKAZAM** — the in-battle coach. A light version runs in your browser; a version that could beat top humans needs a rented cloud computer and months of the AI playing itself.

**Something we got wrong, kept here on purpose.** For two days nobody updated these documents while the code kept changing. The next person to read them — which was us — then described several models completely incorrectly, including calling one "not built yet" when it had been built, tested, and measured. Documents that lag the work don't just go quiet; they actively mislead.

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## Slide 8 — Read more

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Full technical detail, math, and sources: [ABRA white paper](#). Also: [project summary](#) · [technical docs](#) · [model ledger](#).